PLATFORM ARCHITECTURE

High-Level Design (HLD) & Low-Level Design (LLD)

Bridge to Bangladesh

Global Diaspora Engagement Platform

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1. System Overview (High-Level Design)

1.1 Platform Overview

Bridge to Bangladesh is a secure, cloud-based Global Diaspora Engagement Platform designed to connect the worldwide Bangladeshi diaspora community with local organisations, researchers, and civil society actors in Bangladesh. The platform is jointly developed by Revinr IT Ltd on behalf of Oxfam in Bangladesh, and operates as a multi-portal, multi-role digital ecosystem.

The platform is structured around a microservices architecture deployed on leading cloud infrastructure (AWS or Microsoft Azure), providing independent scalability, fault isolation, and long-term extensibility. The domain is bridge2bangladesh.org.

1.2 Purpose of the System

Bridge to Bangladesh is designed to address a clearly identified structural gap: the absence of a trusted, verified, and action-oriented digital platform connecting the global Bangladeshi diaspora with local development actors. The platform's purpose is threefold:

- To build a trusted, verified diaspora community — enabling diaspora members to create structured professional profiles and engage with verified local organisations.
- To facilitate meaningful matchmaking — surfacing connections between diaspora expertise and local NGO/CSO needs that personal networks cannot provide.
- To provide outcome visibility — giving time-poor diaspora professionals transparent evidence that their contributions lead to traceable, measurable impact.

1.3 Key Objectives

Objective	Description
Trust-First Design	Onboarding must lead with Oxfam co-branding, verification signals, and visible governance commitment before requesting profile data. No feature has value to a user who does not trust the platform.
Verified-Before-Visible	No diaspora profile, NGO organisation, campaign, or content item becomes publicly visible without explicit admin approval. Enforced at the service layer, not only at the UI.
Smart Matchmaking	The platform's core value proposition — surfacing connections that personal networks cannot find, using a five-factor weighted recommendation algorithm. Phase 2
Outcome Visibility	Integrated impact dashboards and project tracking are system requirements, not reporting tools. Primary retention mechanism for time-poor professionals.
Political Neutrality	SDG-aligned taxonomy, moderation policy, and profile field design actively prevent political capture. Engineered, not assumed.
Global Accessibility	Not UK-centric. English and Bengali from day one. Designed for users across 12+ countries, multiple timezones, and lower-bandwidth environments.

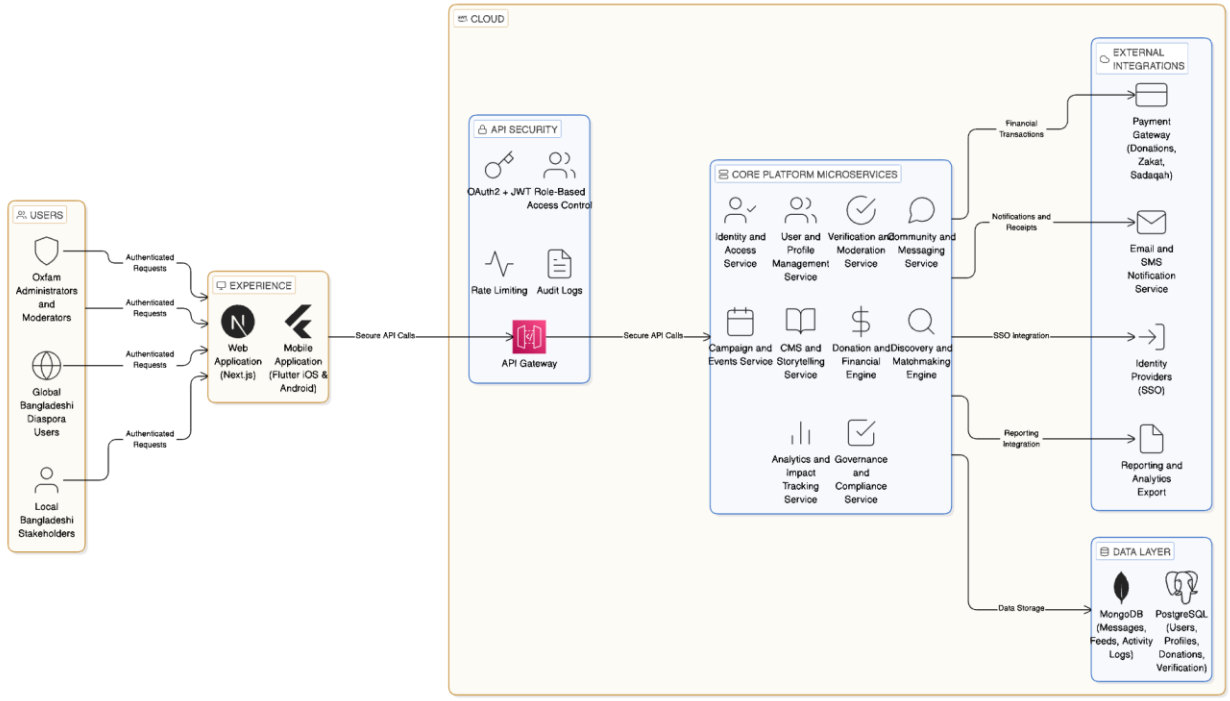
Phase-Based Rollout	Phase 1 focuses exclusively on diaspora user acquisition and community building. Local NGO/CSO onboarding is deferred to Phase 2, contingent on critical mass of 25,000–40,000 active diaspora members.

2. System Architecture (High-Level Design)

2.1 Overall Architecture Description

The Bridge to Bangladesh platform is built as a secure, layered, cloud-based system underpinned by a microservices architecture. This approach was chosen deliberately over a traditional monolithic model: it allows each functional domain to scale independently, contain failures without cascading, and evolve without disrupting other parts of the system.

The architecture is organised into six clearly delineated layers, each with defined responsibilities and secure interfaces to adjacent layers. All data flows between these layers are authenticated, auditable, and GDPR-compliant by design.



Figure— Bridge to Bangladesh: High-Level System Architecture Overview

2.2 Six Architectural Layers

Layer	Description
1. User Layer	Three primary user groups: Oxfam Administrators, Global Bangladeshi Diaspora Users, and Local Stakeholders (NGOs, CSOs, researchers). Each group accesses the system through appropriately scoped, role-controlled interfaces.
2. Experience Layer	Web application (Next.js 14 / Tailwind CSS) and Flutter-based mobile application (iOS and Android).
3. API and Security Layer	The single secure entry point for all backend communications. Implements Zero Trust security — no request is trusted by default. Handles authentication (OAuth2 + JWT), authorisation (RBAC), request validation, rate limiting, and audit logging.
4. Core Platform Microservices	Eleven independent services, each representing a single business domain, owning its own data store, and exposing a defined API. Built on Domain-Driven Design (DDD) principles for loose coupling and high cohesion.
5. External Integrations Layer	Connects to payment gateways, email notification services, SSO providers (LinkedIn, Google), and reporting/analytics tools — all via secure, audited APIs routed through the API Gateway.
6. Data Layer	Hybrid database architecture: PostgreSQL for structured transactional data (ACID-compliant) and MongoDB for high-throughput real-time community and messaging data. Database-per-microservice pattern enforced.

2.3 Major System Components

2.3.1 Eleven Core Microservices

Service	Domain	Key Responsibilities
Identity & Access Service	Authentication	User registration/login; OAuth2+JWT token generation; RBAC enforcement; session management; GDPR consent tracking.
User & Profile Management Service	Profiles	Profile creation/updates; skills, interests, and SDG sector tagging; geographic data; profile visibility controls; multi-role support.
Verification & Moderation Service Phase 2	Trust	Profile verification pipelines; document submission and review; approval/rejection workflows; trust badge assignment. No entity is publicly visible without passing through this service.
Community & Messaging Service Phase 2	Engagement	Secure direct messaging; community feeds; comments and reactions; content posting; content filtering, reporting, flagging; admin intervention tools.
Discovery & Matchmaking Service Phase 2	Discovery	Advanced search by sector, location, interest; five-factor weighted recommendation engine; follow/save profiles; expression of interest; future AI-assisted matching extension.
Campaign & Events Service Phase 2	Campaigns	Campaign landing pages; event scheduling; participation tracking; awareness and advocacy initiatives; SDG-taxonomy-aligned campaigns.

CMS & Storytelling Service	Content	Blogs and articles; video and media uploads; impact stories; content categorisation by SDG sector; moderated user-generated and admin-authored content.
Donation & Financial Service Phase 2	Finance	Donation processing; Zakat/Sadaqah tagging and categorisation; fund allocation; transaction tracking; receipt generation; financial audit logs; payment gateway integration.
Analytics & Impact Tracking Service Phase 2	Analytics	User engagement metrics; campaign performance tracking; donation trends; geographic analytics; SDG contribution dashboards; exportable reports.
Governance & Compliance Service Phase 2	Governance	Admin dashboards; role hierarchy management; audit log access; abuse detection workflows; safeguarding controls; GDPR rights management.
Notification Service	Notifications	Email and in-app push notifications; donation confirmations; verification status updates; campaign alerts; match and connection notifications.

2.4 Component Interaction and Request Flow

Every user-initiated action follows a consistent, secured flow through the system:

Step	Interaction
Step 1	Users initiate requests through the web or mobile application interface.
Step 2	All requests are routed to the API Gateway — the single secure entry point to the entire backend.
Step 3	The Gateway validates the JWT token, enforces RBAC, checks rate limits, and logs the request to the audit trail.
Step 4	Validated requests are routed to the appropriate microservice based on the endpoint.
Step 5	The microservice executes business logic and reads from or writes to its designated database.
Step 6	External integrations (payment gateway, email, SSO) are called when the action requires them.
Step 7	The processed response is returned to the user through the Gateway.
Step 8	All actions generate an immutable audit log entry for compliance and governance review.

3. Technology Stack

The technology stack has been selected to balance technical excellence, operational sustainability, and long-term maintainability. Every choice is grounded in three criteria: serving the platform's security and scalability requirements; being supportable by the Oxfam partner team and successor vendors; and having a strong, active open-source or commercial ecosystem.

3.1 Frontend Technologies

Technology	Details & Rationale
Next.js 14 (Web — Primary)	Server-side rendering for performance and SEO; fully responsive layout across desktop, tablet, and mobile; WCAG 2.1 AA compliant; strong TypeScript ecosystem; widely supported and maintainable.
Tailwind CSS	Utility-first CSS framework used with Next.js 14 for the web frontend.
Flutter / React Native (Mobile)	Single codebase for iOS and Android; feature parity with web; optimised for lower-bandwidth environments; growing ecosystem with strong Google backing.
Languages	English and Bengali (Unicode) from day one across both web and mobile interfaces.

3.2 Backend Technologies

Technology	Details & Rationale
Node.js (Express)	High-performance, scalable microservice development. Used for event-driven, high-concurrency services such as messaging and notifications.
Python (FastAPI)	Used for data-intensive services such as analytics and matchmaking due to async performance and the Python data ecosystem.
gRPC (Internal)	High-performance internal service-to-service calls with strong typing and low latency.
REST APIs (External)	Used for API Gateway and all external-facing integrations.
RabbitMQ / Bull MQ	Asynchronous task queues for notifications and email processing between services.
Redis (Caching)	In-memory caching for high-frequency, low-change read operations: profile lookups, match scores, SDG taxonomy. Reduces database load and improves API response times.
OAuth2 + JWT	Industry-standard stateless authentication. Supports login via LinkedIn and Google. Enables multi-device session handling and role claim embedding in tokens.

3.3 Database

Database	Details & Rationale
PostgreSQL (Relational)	ACID-compliant relational database for all structured, transactional data: identity, profiles, verification, donations, campaigns, governance, analytics. Read replica capability and GDPR-ready data management tools.
MongoDB (NoSQL — Atlas)	Flexible document schema for messaging and feed data. High write throughput for real-time community features. Horizontal sharding support for future scale. Single cluster.

3.4 Hosting & Infrastructure

Component	Technology & Detail
Cloud Provider	Amazon Web Services (AWS) or Microsoft Azure — global availability zones for low-latency access across UK, North America, Europe, and Asia-Pacific.
Containerization	Docker — all microservices packaged as Docker containers. Each service runs independently in its own container.
CI/CD Pipeline	GitHub Actions — fully automated build-test-deploy pipeline. Docker image creation and push to container registry (ECR/ACR) with health-checked rollouts.
Object Storage	AWS S3 / Azure Blob Storage — scalable storage for user-uploaded verification documents, CMS media assets, and data exports.
Monitoring	Prometheus + Grafana — real-time metrics collection and visual dashboard rendering. Custom alerting rules for performance thresholds and error rate spikes.
Logging	ELK Stack (Elasticsearch, Logstash, Kibana) — centralised log aggregation across all microservices. Full-text search across audit logs.
High Availability	Multi-zone deployment with redundant service replicas and automatic failover. No single point of failure in the production architecture.
Environment Strategy	Three environments: Development, Staging (pre-production/UAT), and Production.

3.5 Third-Party Services & Integrations

Service	Technology & Detail
Payment Gateway	Stripe or equivalent PCI-compliant provider — secure multi-currency payment processing, Zakat/Sadaqah fund categorisation, webhook-based confirmation and receipt generation. PCI-DSS compliant.
Email Notifications	SendGrid / Twilio or equivalent — transactional email delivery with global reach, delivery tracking, bounce management, and branded communication templates.

LinkedIn	LinkedIn OAuth2 — frictionless one-click registration for professional diaspora users; credential verification through LinkedIn's identity infrastructure.
Google	Google OAuth2 — alternative frictionless pathway, particularly for second and third-generation diaspora members.
Reporting & Analytics Export	Export of platform-wide SDG impact data, campaign performance metrics, and donation summaries to external dashboards. Used for Oxfam institutional and donor reporting.

4. Data Flow Architecture

4.1 Diaspora Member Portal — End-to-End Data Flow

The Diaspora Member journey is structured across four sequential phases, each representing a distinct data flow stage:

Phase 1 — Platform Joining (Registration & Authentication)

Flow Step	Data & Action
Route A: Invitation-Based	User receives personalised email from Oxfam → clicks link → submits name, country of residence, password → account activated immediately, no queue.
Route B: Self-Registration	User visits bridge2bangladesh.org → clicks Join as Diaspora → submits name, email, country → account queued for Oxfam admin review (within 24 hours).
Login	User authenticates via email/password or LinkedIn/Google SSO → JWT token issued by Identity Service → RBAC role claims embedded.

Phase 2 — Profile Creation (Four-Step Structured Input)

Step	Data Captured
Step 1 — Identity	Full name, country of residence, professional role.
Step 2 — Interest	SDG sector selections (Health, Education, Climate, Innovation, etc.).
Step 3 — Capability	Skills and expertise taxonomy entries (e.g., surgeon, architect, policy advisor).
Step 4 — Availability	Time windows for engagement; preferred collaboration mode (in-person, online, or hybrid).

Phase 3 — Discovery & Matchmaking

Once a diaspora member sets their availability, their profile is made visible in the directory. NGOs can filter by date, skills, sector, and collaboration type. The Matchmaking Service applies a five-factor weighted algorithm to compute match scores. Connection requests flow from NGO → Diaspora Member → Notification Service → Accept/Decline → Secure messaging channel opened.

Phase 4 — Active Collaboration

After connection is established, data flows through: secure private messaging (Community & Messaging Service), project documentation and tracking (shared workspace), event participation or facilitation, and contribution logging (fed to Analytics & Impact Tracking Service and Gamification Engine).

4.2 Local NGO / CSO Portal — Data Flow (Phase 2)

Phase	Data Flow
Registration	Organisation name, type, registration number, contact person, website → document upload (NGO certificate + government-issued ID) → transmitted to secure document store → placed in Oxfam admin verification queue.
Verification	Admin reviews queue → approves/rejects/requests additional information → upon approval: Verified Badge assigned → profile becomes publicly visible in NGO directory.
Profile Building	SDG sector tags, active project descriptions, team profiles, media assets, collaboration type statements → data feeds into Matchmaking Service to generate match scores.
Matchmaking	NGO filters diaspora directory by sector, skill, country, district of origin, collaboration type → reviews match scores → sends connection request with project description → upon acceptance: secure messaging and shared workspace activated.

4.3 Admin Portal — Governance Data Flow

Workflow	Data Flow
Content Moderation	All user-generated content queued for review → admin approves/edits/removes/restricts → all moderation actions logged in Governance & Compliance Service audit trail.
Invitation Management	Admin sends personalised invitation emails → invitee receives link → completes short registration → account activated immediately (no queue).

4.4 Key Data Inputs and Outputs

Category	Detail
Primary Data Inputs	User registration data; profile attributes (skills, SDG interests, availability); document uploads (verification); user-generated content (posts, articles, stories); donation transactions; campaign and event data.
Primary Data Outputs	Verified user profiles; match scores and recommendations; notification events; donation receipts; impact dashboards; exportable SDG reports; audit logs.
Cross-Service Events (Async)	Donation confirmed → Notification Service (receipt email); Verification approved → Profile Service (verified badge); User activity → Analytics Service (engagement tracking).
UTC Timestamps	All timestamps stored in UTC throughout the system to support accurate cross-timezone display for users across 12+ countries.

5. Integration Points

5.1 External Systems and APIs

The platform connects with four categories of external systems, all via secure, audited APIs. No external integration can access internal services directly; all calls pass through the API Gateway and are logged.

Integration	Purpose and Detail
Payment Gateway (Stripe or PCI-equivalent)	Secure processing of general donations, cause-based appeals, and faith-based giving (Zakat/Sadaqah). Multi-currency support for global diaspora across UK, North America, Europe, and Asia-Pacific. PCI-ready architecture. Receipt and acknowledgment generation routed through Notification Service.
Email — SendGrid / Twilio (or equivalent)	Transactional notifications: donation confirmations, verification status updates, campaign launch alerts, new match notifications, and event reminders. Global delivery capability with delivery tracking and bounce management.
LinkedIn OAuth2 (SSO)	Optional Single Sign-On via LinkedIn OAuth2. Enables frictionless, one-click registration for time-poor professional diaspora users — directly addressing the five-minute registration requirement.
Google OAuth2 (SSO)	Optional Single Sign-On via Google OAuth2. Alternative frictionless registration pathway for second and third-generation diaspora members.
Reporting & Analytics Export	Export of platform-wide SDG impact data, campaign performance metrics, and donation summaries to external dashboards or reporting tools. Used for Oxfam institutional reporting, donor transparency disclosure, and government engagement updates.
Community & Messaging Service → Governance Service	Content flagging and moderation reports are routed from the Community & Messaging Service to the Governance & Compliance Service's moderation workflow.

5.2 Data Exchange Mechanisms

Pattern	Technology & Use Cases
Synchronous (Request-Response) — REST	Used for API Gateway-to-microservice routing and all external-facing communications.
Synchronous (Request-Response) — gRPC	Used for high-performance internal service-to-service calls where speed and efficiency are critical (e.g., identity validation during profile retrieval).
Asynchronous (Event-Driven) — RabbitMQ / BullMQ	Used for task queues: notification triggers, email processing. Used for high-throughput event streaming: activity logs, analytics pipeline, donation event chains.

6. Low-Level Design (LLD)

6.1 Database Design — Hybrid Architecture

The platform employs a database-per-microservice pattern. No service ever directly accesses another service's database. All cross-service data exchange occurs exclusively through API calls.

Database	Assigned Services / Domains
PostgreSQL (Relational — ACID)	Identity and access; user profiles; verification records; matchmaking connections; donations and financial transactions; campaign and CMS content; governance and moderation; analytics and impact metrics.
MongoDB (NoSQL — Atlas)	Real-time messaging; community posts and feeds; comments and reactions; activity logs.

6.2 Microservice Database Schemas

6.2.1 Identity Service — PostgreSQL

Handles authentication, authorisation, session management, and GDPR compliance tracking. All users, regardless of role, have a record in this database.

Table	Key Fields	Description
users	id (UUID, PK), email (unique), password_hash, status, created_at, updated_at	Core user record. Status: active | suspended.
roles	id, name	Role types: diaspora | local_user | admin | moderator.
user_roles	user_id (FK), role_id (FK)	Many-to-many link: one user may hold multiple roles.
sessions	user_id, refresh_token, expires_at	JWT refresh token management and session expiry.
consents	user_id, consent_type, accepted_at	GDPR compliance record. Types: GDPR | marketing.

6.2.2 Profile Service — PostgreSQL

Stores structured profiles for diaspora members and local stakeholders. Powers the matchmaking algorithm and governs what each user sees about others.

Table	Key Fields	Description
profiles	id (UUID), user_id (FK), profile_type, full_name, country, city, bio, verified_status, created_at	Profile types: diaspora | entrepreneur | student | CSO. verified_status controlled by Verification Service.

skills	id, name	Skill taxonomy entries used for matchmaking filters.
profile_skills	profile_id, skill_id	Many-to-many: links profiles to declared skills.
organizations	id, name, type	Organisation types: NGO | startup | academic.

6.2.3 Campaign and CMS Service — PostgreSQL

Table	Key Fields	Description
campaigns	id, title, description, category, start_date, end_date	Category = SDG sector tag.
events	id, campaign_id, title, event_date	Event linked to parent campaign.
content	id, title, body, type, published_by	Types: blog | story | video. published_by = author/admin who published.

6.2.4 Governance and Moderation Service — PostgreSQL

Table	Key Fields	Description
reports	id, reported_by, content_id, reason, status	Flagged content or user report with reason and resolution status.
moderation_actions	id, moderator_id, action, target_id, timestamp	Full admin action audit trail: what was done, to what, and when.

6.3 Database Design Principles

Principle	Implementation
Data Isolation	Each microservice exclusively owns its database. No cross-database joins. All cross-service data exchange occurs via authenticated API calls only.
GDPR Compliance	Dedicated consents table in Identity Service. Data deletion and portability workflows built at schema level. Right to erasure enforced via cascading delete procedures. Data minimisation: only data essential to platform function is collected and retained.
Financial Audit	Immutable transaction history with payment gateway references. Separate donations and transactions tables for complete audit trail. Restricted_flag on funds table enforces designated giving compliance.
RBAC Enforcement	Role and user_roles tables in Identity Service provide the authority record. Permissions enforced at API Gateway level using JWT role claims — not only at the UI layer.
Event-Driven Readiness	Architecture supports Apache Kafka integration for real-time event streaming. Donation confirmation events, user activity tracking, and notification trigger pipelines can be added without schema changes.

Scalability	PostgreSQL read replicas for high-traffic read-heavy services. Sharding-ready MongoDB configuration. Redis caching layer for frequent, low-change queries. Horizontal scaling via cloud infrastructure auto-scaling.
Security	Encryption at rest via AWS RDS encryption and MongoDB Atlas. TLS enforced in transit. Row-level security policies in PostgreSQL where applicable. Audit logs for all financial and admin actions. Configurable data retention policies per service.

6.4 API and Security Layer — Detailed Specification

Responsibility	Implementation
Authentication	OAuth2 + JWT: stateless token-based authentication issued by the Identity Service. Tokens carry role and permission claims, expire on schedule, and support refresh and multi-device session handling.
Authorisation (RBAC)	Five role types: Diaspora User, Local Stakeholder, Verified Organisation, Moderator, Oxfam Administrator. Each with precisely defined permissions enforced at the API Gateway level — not only in the UI.
Request Validation	Input data format checks, required field verification, file upload validation, and payload size limits applied to every incoming request before it reaches any microservice.
Data Protection	Encryption in transit: HTTPS/TLS 1.3. Encryption at rest: AES-256 (PostgreSQL via RDS; MongoDB via Atlas). Sensitive fields masked in logs and API responses.
Rate Limiting	Per-user and per-IP rate limiting; throttling of excessive requests; IP blocking for suspicious or malicious behaviour; optional CAPTCHA on public-facing endpoints.
Audit Logging	All user login/logout events, profile updates, donation transactions, admin actions, and content moderation activities logged centrally and immutably.
Inter-Service Security	Service-to-service communication authenticated via gRPC/REST over TLS with network-level isolation. Internal services cannot communicate without authorisation.

6.5 Scalability and Future Readiness

Dimension	Design Approach
Horizontal Scaling	Cloud infrastructure supports independent scaling of each microservice under variable load. The messaging service scales during community events; the donation service scales during emergency appeals — each without affecting other components or requiring coordinated downtime.
Future Extensibility	Modular microservices architecture allows new services to be added (e.g., investment facilitation, mentorship calendar integration, government liaison portal) without disrupting existing functionality. New payment providers, geographic regions, and institutional partners can be integrated via the existing API layer.
AI & Analytics Roadmap	The Discovery and Matchmaking Service is architected to accept an AI-assisted matching layer in a future phase, replacing the initial rule-based algorithm with a machine learning model once sufficient engagement data has been collected. The Analytics Service is data warehouse-ready for advanced impact modelling.

7. System Constraints & Assumptions

7.1 Scalability Constraints

- The platform must support concurrent global access from users across 12+ countries and multiple time zones simultaneously without performance degradation.
- Architecture must accommodate a minimum of 10,000 registered users at launch, scaling to 25,000–40,000 active users (Phase 1 critical mass target) and 100,000+ within 24 months without architectural re-engineering.
- Database schema and API design must support addition of new user types, content categories, and Phase 2 service modules without breaking Phase 1 functionality.

7.2 Security Constraints

- Full compliance with OWASP Top 10 security standards throughout development and post-deployment — confirmed as non-negotiable.
- Multi-factor authentication (MFA) mandatory for all user and admin account types.
- Data encryption in transit (TLS 1.3) and at rest (AES-256) for all user data.
- Role-based access control enforced at API level, not only at the user interface level.
- Regular penetration testing and cybersecurity audit by a certified third-party expert prior to launch and at defined intervals post-launch.
- All coding and technical standards compliance embedded from development commencement, not retrofitted.

7.3 Performance Constraints

- Page load time under 3 seconds on standard 4G mobile connection.
- API response time under 500 milliseconds for standard queries.
- 99.5% uptime SLA with documented disaster recovery and backup protocols.
- UTC-stored timestamps throughout the system to support accurate cross-timezone display for users across 12+ countries.

7.4 Accessibility Constraints

- Full compliance with WCAG 2.1 AA accessibility standards throughout the web platform.
- Screen-reader compatibility for all core interface elements.
- Platform optimised for lower-bandwidth environments given global diaspora user base and Bangladesh connectivity considerations.

7.5 Data Privacy & GDPR Constraints

- Full compliance with UK GDPR and applicable Bangladesh data protection regulations — confirmed as non-negotiable.
- Granular user consent management from registration and throughout account lifecycle.
- Right to erasure: users can request deletion of personal data with automated compliance workflow.

- Data portability: users can export personal data in standard formats.
- No third-party data sharing without explicit user consent.
- GDPR rights requests (access, erasure, portability) tracked and processed within statutory timeframes via the Governance Service GDPR management module.

7.6 Phase Sequencing Assumption

Local NGO/CSO onboarding is deferred to Phase 2. Phase 1 focuses exclusively on diaspora user acquisition and community building. Local organisation activation is contingent upon the diaspora user base reaching a critical mass of 25,000–40,000 active members. This is a strategic sequencing decision, not a rejection of local stakeholder inclusion.

8. User Journeys and Data Flow Diagram

The user journey diagrams presented in this section translate the data flow architecture into lived user experience. Each journey traces the complete path of a user type from initial platform entry through to active collaboration and ongoing engagement. The journeys are grounded in the stakeholder consultation findings particularly the emphasis on minimum friction, maximum trust, and structured engagement pathways and operationalise the functional requirements defined in the FRS. The flow diagrams for each persona served as the design starting point for both the data flow architecture and the functional requirements, ensuring full traceability from stakeholder insight to system specification.

8.1 Diaspora Member Portal Journey

The Diaspora Member journey is designed around two non-negotiable principles drawn directly from stakeholder consultation: minimum friction for time-poor professionals, and maximum trust through verified, structured engagement. The journey is structured across four sequential phases.

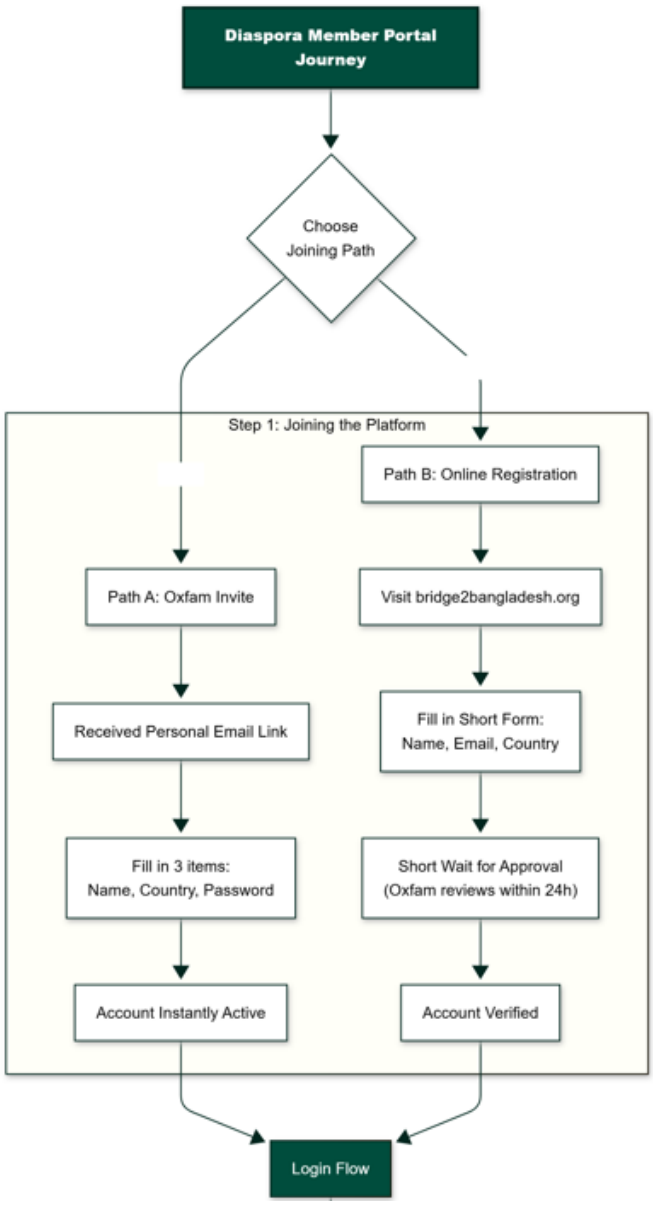
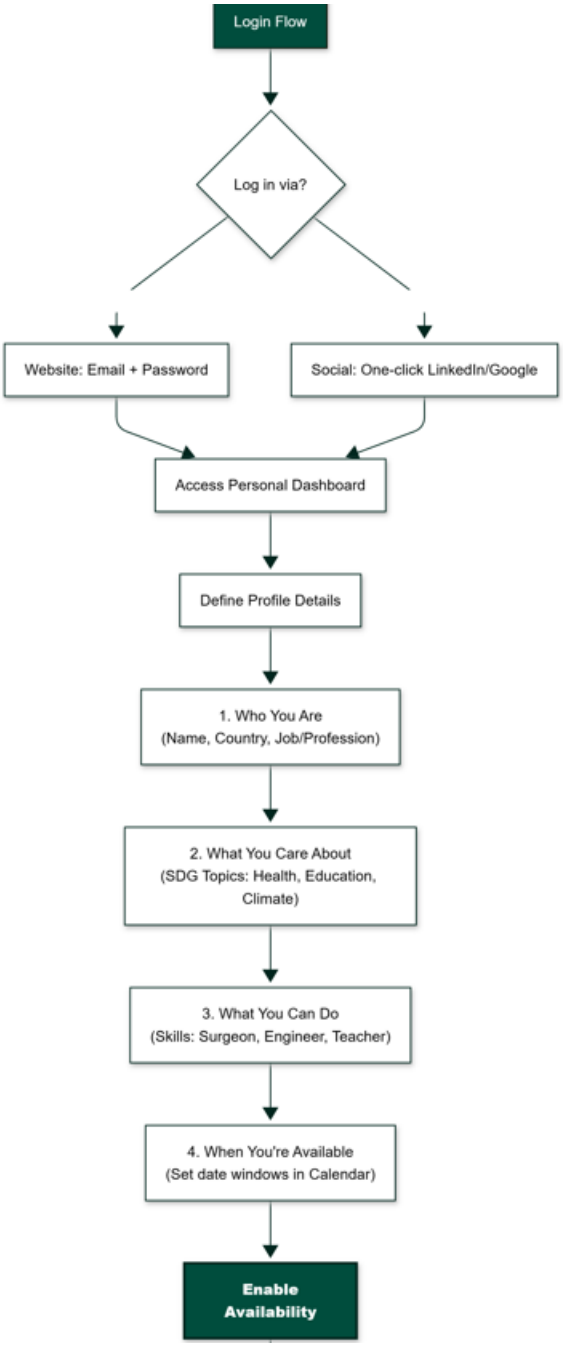


Figure — Diaspora Member Portal Journey: End-to-End Interaction Flow



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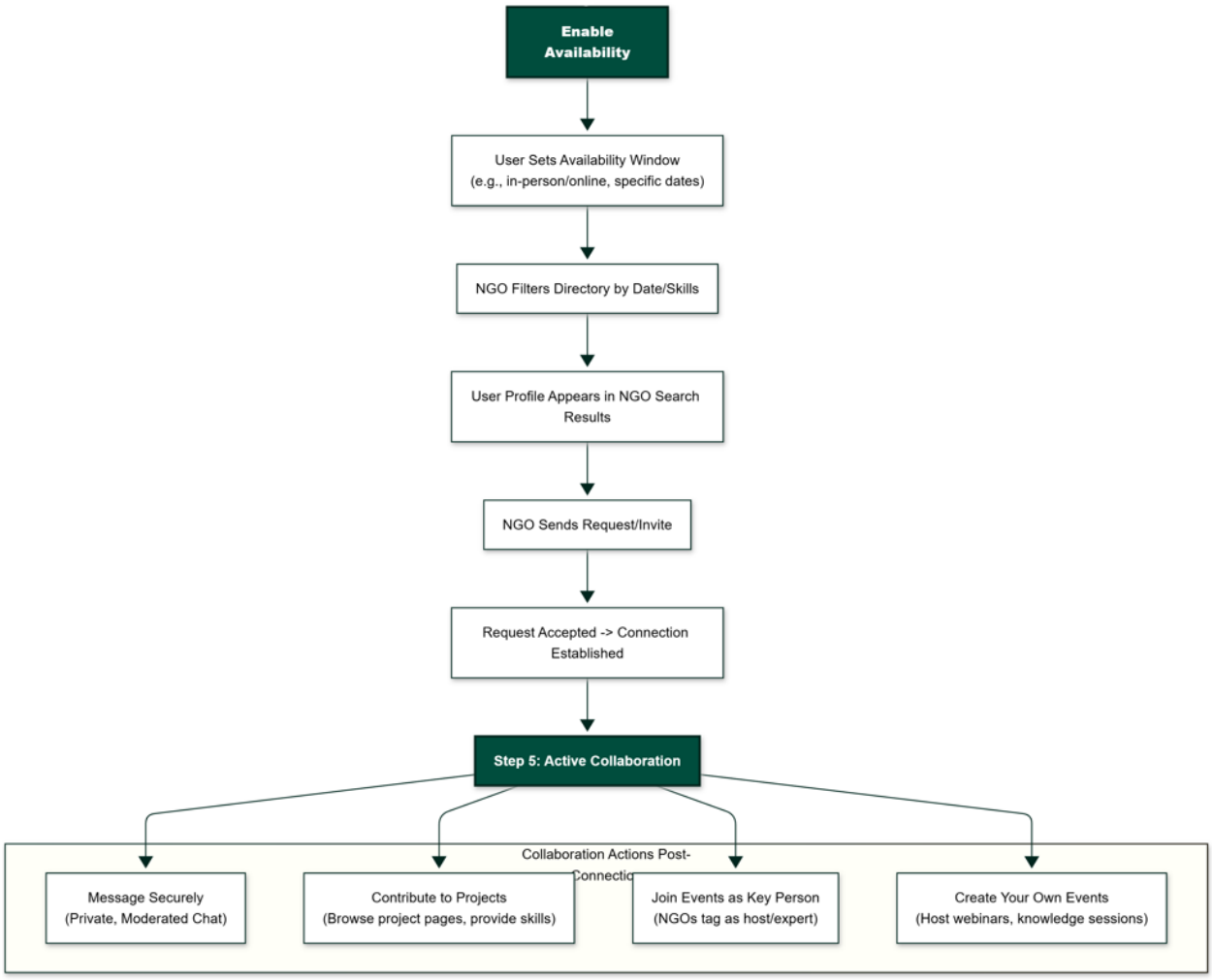


Figure — Diaspora Member Portal Journey: End-to-End Interaction Flow

The Diaspora Member journey is designed around two non-negotiable principles drawn directly from stakeholder consultation: minimum friction for time-poor professionals, and maximum trust through verified, structured engagement. The journey is structured across four sequential phases:

Phase 1 — Joining the Platform: Diaspora members can join the platform through two registration routes. The first route is invitation-based. In this case, the user receives a personal email from Oxfam, clicks a personalised link, enters three basic details such as name, country of residence, and password, and the account becomes active immediately. This route is intended for selected early users identified through Oxfam's networks. The second route is open self-registration. Here, the user visits **bridge2bangladesh.org**, clicks **Join as Diaspora**, completes a short form with name, email, and country, and then waits for Oxfam to review and approve the account, usually within 24 hours. After this, both routes lead to the same login process, where users can sign in using email and password or through LinkedIn or Google single sign-on. This keeps the onboarding process simple and aligned with the five-minute registration target.

Phase 2 — Profile Creation and Definition:

After login, users build a structured profile across four key dimensions:

Identity: name, country, profession

Interest: SDG sectors (e.g., Health, Climate, Education)

Capability: skills and expertise

Availability: time windows for engagement

This profile acts as the foundation for matchmaking, ensuring that all future interactions are relevant and purpose driven.

Upon first login, users are guided through a simple four-step profile setup that helps the platform understand who they are and what kind of opportunities may be relevant to them. The first step captures basic identity information such as name, country, and professional role. The second step focuses on areas of interest, where users select the SDG themes they care about, such as Health, Education, Climate, Innovation, and related sectors. The third step records what the user can contribute by listing skills and expertise, for example surgeon, architect, engineer, teacher, or policy advisor. The fourth step captures availability, allowing users to indicate when they are free to engage and whether they prefer in-person, online, or hybrid collaboration. Together, these four parts create a structured profile that supports the platform's matching process and helps surface opportunities that are relevant, practical, and timely, rather than broad or generic.

Phase 3 — Enabling Availability and Connecting: Once a diaspora member sets their availability, the profile becomes visible to NGOs searching the directory by date, skills, sector, and type of collaboration. If an NGO finds a suitable match, it can send a connection request explaining the purpose of the project or engagement. The diaspora member then receives a notification and can either accept or decline the request. This approach helps ensure that connection requests are relevant and purposeful rather than random or speculative. It also supports trust by making the reason for contact clear from the beginning. When the request is accepted, a secure messaging channel is opened so that both parties can communicate directly within the platform.

Phase 4 — Active Collaboration: After a connection is established, diaspora members can engage in several structured ways through the platform. They can communicate through a secure private messaging channel, where conversations remain within the platform and are subject to moderation protocols for safety and accountability. They can also contribute directly to specific NGO projects by reviewing project pages, offering skills-based support, following documentation, and tracking progress over time. In addition, connected diaspora members may take part in events as key contributors, for example as hosts, speakers, or facilitators in webinars, workshops, or knowledge-sharing sessions. More experienced members may also create and host their own events, such as mentorship sessions, professional discussions, or learning-based webinars, subject to admin approval. This allows the platform to support not only one-to-one collaboration but also wider community engagement and knowledge exchange.

Journey Outcome

The diaspora journey transforms:

Intent → Verified Connection → Structured Collaboration → Measurable Impact

> *"The platform must supplement what personal networks cannot do: surface new, verified connections across sectors and geographies that would otherwise never form."*
>
> **— B2B Stakeholder Survey Analysis, March 2026**

8.2 Local NGO / CSO Portal Journey

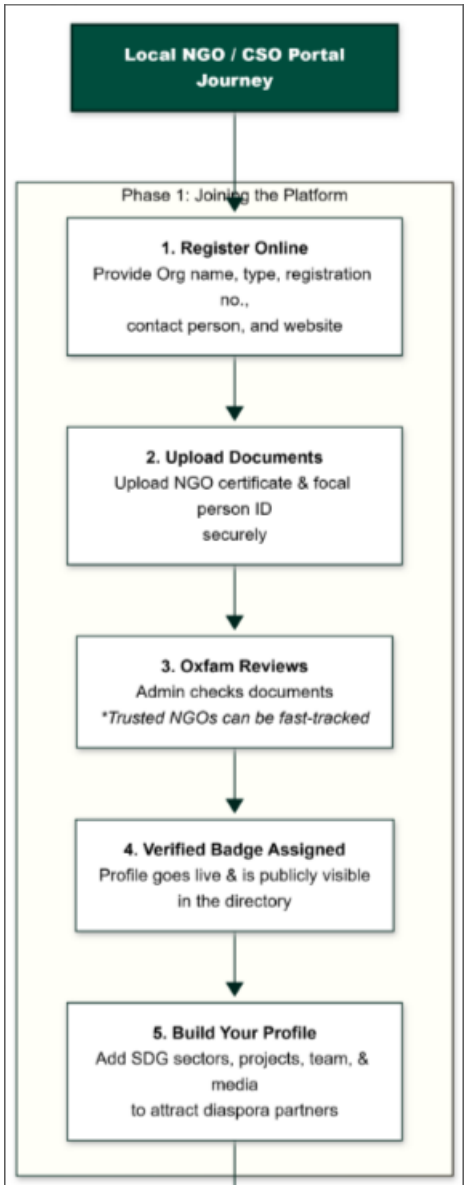
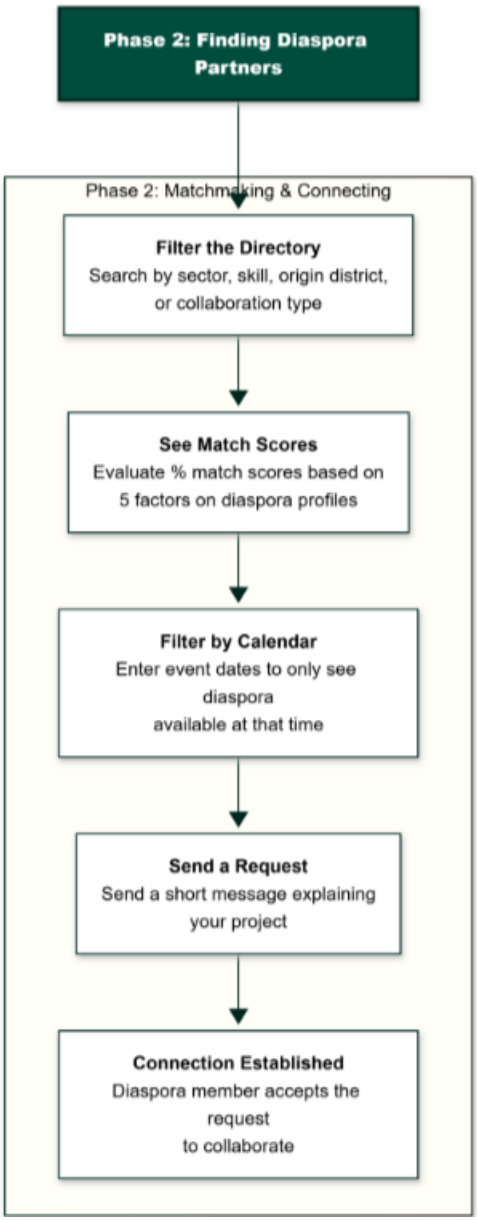


Figure — Local NGO / CSO Portal Journey: Registration to Active Partnership



Figure— Local NGO / CSO Portal Journey: Registration to Active Partnership

The Local NGO journey is structured to give Bangladesh-based organisations a credible, systematic pathway to diaspora partnership while ensuring that the platform's verification standards are never compromised. The journey is sequenced so that organisational credibility is established before any diaspora connection is possible — directly operationalising the trust-first principle identified across all stakeholder consultations.

Phase 1 — Joining the Platform: Registration begins with the organisation providing its name, organisation type, registration number, contact person, and official website. This baseline data is used to initiate the verification process. The organisation then securely uploads two key documents: its NGO certificate (or equivalent legal registration) and the focal person's government-issued ID. These documents are transmitted to a secure document store and placed in the Oxfam admin verification queue. Organisations with an existing

Oxfam partnership relationship can be fast-tracked through the queue without requiring the standard review cycle. Upon admin approval, a Verified Badge is assigned and the organisation's profile becomes publicly visible in the NGO directory — making this the first moment at which any diaspora user can discover the organisation.

Building the Organisational Profile: Once verified, NGOs build out a rich, structured profile to maximise their discoverability and appeal to diaspora partners. Profile fields include SDG sector tags (aligning the organisation's work to the platform's taxonomy), active and upcoming project descriptions, team profiles, media assets (photos, videos, impact stories), and a statement of the types of collaboration being sought (mentorship, knowledge transfer, co-design, volunteering, funding). This profile functions as the organisation's primary interface with the global diaspora community and directly influences their match scores.

Phase 2 — Finding Diaspora Partners: Verified NGOs access the matchmaking tools to actively identify and engage diaspora partners. The directory can be filtered by sector, skill, country of residence, district of origin in Bangladesh, and collaboration type. The platform displays match scores — computed from five weighted factors — for each diaspora profile, enabling NGOs to prioritise the most relevant connections. An availability calendar filter allows NGOs to identify diaspora members available within a specific date window, critical for time-sensitive project needs. Having identified a suitable partner, the NGO sends a brief, purpose-described connection request explaining the project, the specific input sought, and the expected time commitment. When the diaspora member accepts, the connection is established and both parties gain access to the secure messaging channel and shared project workspace.

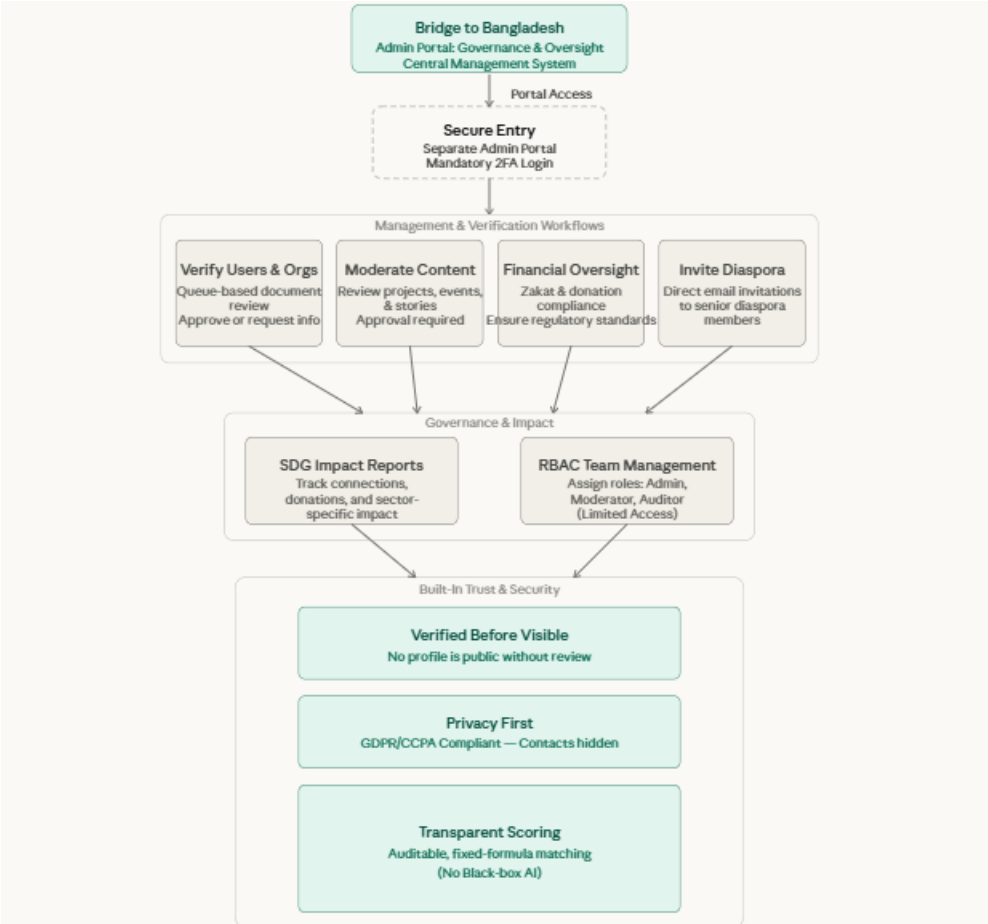
Journey Outcome

The NGO journey transforms:

Local Need → Verified Visibility → Targeted Matching → Structured Partnership

> *"Verification is critical — we can copy this approach. Track and share impact — people want to see results. Add community features to keep people engaged."*
>
> **— Platform Comparative Analysis — Lessons from Go for Change and African Diaspora Network**

8.3 Admin Portal Journey



Figure— Admin Portal Journey: Governance, Verification, and Oversight Workflow

The Admin Portal operationalises Oxfam's institutional governance mandate for the platform. It is accessed exclusively through separate, mandatory two-factor authentication (2FA) protected login, isolating administrative functions from the general user environment and ensuring that all platform governance actions are performed by authorised, authenticated personnel. The admin journey is structured around five integrated management workflows that together constitute the platform's complete governance operation:

Verify Users and Organisations: The Verification Queue is the most critical governance function. All diaspora member accounts and NGO/CSO organisation profiles requiring verification are surfaced here with their submitted documentation. Administrators conduct a queue-based document review, assessing submitted registration certificates, governance records, and ID documents against the platform's verification criteria. They can approve, reject, or request additional information from the applicant. No profile — individual or organisational — becomes publicly visible on the platform without an explicit approval action from an administrator. This is the primary mechanism through which the platform's trust architecture is operationally enforced and sustained over time.

Moderate Content: The content moderation workflow ensures the platform maintains its commitments to political neutrality, safeguarding standards, and content quality. All projects, events, campaigns, community posts, and editorial content submitted by users are queued for moderation review before publication. Administrators can approve content as submitted, request edits, edit content directly, remove content that

violates platform policies, or restrict accounts that repeatedly breach moderation standards. The moderation module includes automated flagging for predefined keyword and content patterns, surfacing high-risk content for priority review. All moderation actions are logged in the immutable audit trail.

Financial Oversight: The Finance Dashboard provides Oxfam with real-time, granular visibility into all financial flows on the platform — including general donations, cause-based appeals, and faith-based giving (Zakat and Sadaqah). Every transaction is logged with its full payment gateway reference, fund allocation designation, compliance status, and donor details. Administrators can monitor fundraising campaign performance, identify anomalies, generate financial compliance reports, and ensure that all donations reach their designated funds in accordance with Oxfam's financial policies and applicable regulatory requirements. This dashboard is also the primary tool for providing financial transparency reporting to donors and institutional partners.

Invite Diaspora: Administrators have direct access to an invitation management tool, enabling them to send personalised email invitations to senior or strategically important diaspora members identified through Oxfam's global networks. This curated invitation pathway operates in parallel with the open self-registration flow, allowing Oxfam to build a high-quality, credible initial user base of influential professionals who might not proactively discover the platform through organic channels. Each invitee receives a personalised link that bypasses the standard approval wait and activates their account immediately upon completion of the short registration form, providing a premium onboarding experience for priority members.

SDG Impact Reports and RBAC Team Management: The SDG Impact Reports module gives administrators a real-time overview of how the platform is contributing across the Sustainable Development Goals. Impact can be viewed by type of connection, sector, geography, and time period, allowing Oxfam to generate meaningful reports for donors, institutional partners, and relevant government stakeholders. Alongside this, the RBAC Team Management function allows senior administrators to assign different roles within the governance team, such as Moderator or Auditor, each with clearly defined permissions. This helps the platform grow in a controlled way while maintaining security, accountability, and effective oversight.

Journey Outcome

The admin journey ensures:

Trust → Verification → Governance → Transparency → Platform Sustainability

9. Summary

Bridge to Bangladesh is a secure, cloud-native, microservices-based diaspora engagement platform developed by Revinr IT Ltd for Oxfam in Bangladesh. The platform is designed to connect the global Bangladeshi diaspora with verified local organisations, researchers, and civil society actors through trust-first design, smart matchmaking, and transparent impact tracking.

The architecture is organised into six layers — User, Experience, API/Security, Microservices, External Integrations, and Data — with eleven independently deployable microservices built on Domain-Driven Design principles. The technology stack comprises Next.js 14 and Flutter for frontend delivery; Node.js/Express and Python/FastAPI for backend microservices; PostgreSQL and MongoDB for hybrid data storage; Redis for caching; RabbitMQ and BullMQ for asynchronous messaging; and Docker with GitHub Actions CI/CD for deployment on AWS or Microsoft Azure.

Security is foundational, not supplementary: the platform enforces a Zero Trust security model, TLS 1.3 encryption in transit, AES-256 encryption at rest, OAuth2 + JWT authentication, five-role RBAC enforced at the API Gateway level, and comprehensive immutable audit logging. OWASP Top 10 compliance, GDPR adherence, WCAG 2.1 AA accessibility, and MFA for all user types are non-negotiable requirements confirmed by all key stakeholders.

The database architecture follows a strict database-per-microservice isolation pattern, with PostgreSQL managing all structured transactional data and MongoDB managing high-throughput real-time community and messaging data.

The platform is designed for phased delivery: Phase 1 delivers the full diaspora-facing portal, community features, content hub, and gamification engine. Phase 2 introduces local NGO/CSO onboarding, the full matchmaking module, donation infrastructure, and mentorship functionality — contingent on reaching a diaspora critical mass. The architecture is explicitly designed to accommodate this phase-based extensibility without requiring architectural re-engineering.

Architecture Dimension	Key Decision / Value
Architectural Pattern	Microservices — 11 independent services, each owning its own database and API.
Security Model	Zero Trust at API Gateway; AES-256 at rest; TLS 1.3 in transit; RBAC enforced at API level.
Database Strategy	Hybrid PostgreSQL (relational/transactional) + MongoDB (NoSQL/real-time); database-per-microservice isolation.
Frontend	Next.js 14 (web, SSR, WCAG 2.1 AA); Flutter (iOS/Android, feature parity).
Deployment	Docker containerisation; GitHub Actions CI/CD; AWS or Azure; multi-zone high availability.
Compliance	OWASP Top 10; UK GDPR; CCPA; WCAG 2.1 AA; PCI-ready payment architecture.
Scalability Target	10,000 registered users at launch → 100,000+ within 24 months without re-engineering.
Inter-Service Comms	REST (external) + gRPC (internal sync) + RabbitMQ/Kafka (async event-driven).
Governance Model	Verified-before-visible; 2FA-protected Admin Portal; immutable audit trail; RBAC team management.
Phase 2 Readiness	Modular design supports NGO onboarding, AI-assisted matchmaking, and data warehouse upgrade without disruption to Phase 1.